MagicLoader Implementation Guide

for Windows® CE

MMSP™ 2

User Manual

MDMP2520FMLUM-WC

Ver. 2.2 / October 27, 2005

MP2520F

MagicLoader Implementation guide for Windows® CE.NET for MMSP 2 Application Processor

MAGICEYES

Important Notice

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Document History

Version	History	Date
1.0	MagicLoader Implementation guide	2005. 9. 8
2.0	-Add file download function from PC to NAND flash thru USB -Add Logo graphic file display function	2005.10.27
2.1	-Add Make logo graphic file	2005.11.2
2.2	-Add Display device configuration	2006.3.8

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1. MagicLoader

1.1. What is the MagicLoader

Generally, MP2520F needs two bootloader. NANDBOOT.bin is the first bootloader for reading the second bootloader. MagicLoader.bin(or EBOOT.nb0) is the second bootloader for reading the OS image file. Boot sequences of MP2520F are as follows :

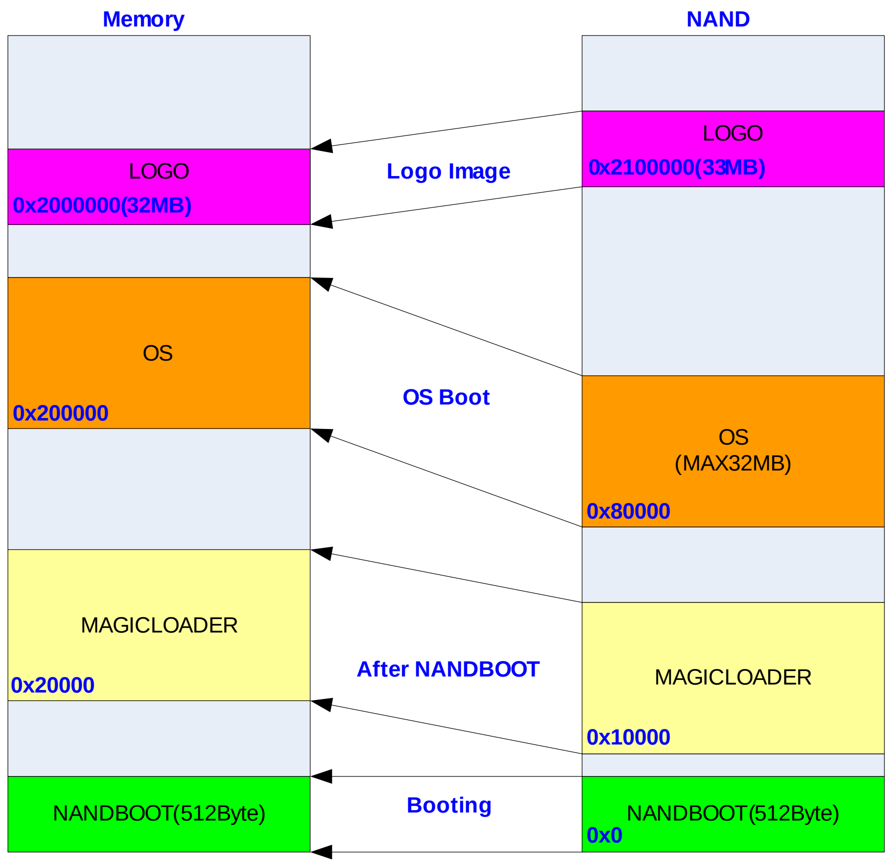
(1) When reset signal is asserted to MP2520F, MP2520F copies the first bootloader(NANDBOOT.bin) from NAND flash to SDRAM address 0

(2) NANDBOOT.bin is executed (MagicLoader.bin(or EBOOT.nb0) is copied from NAND flash to SDRAM address 0x20000)

(3) MagicLoader.bin(or EBOOT.nb0) is executed (OS image file is copied from NAND flash to SDRAM address 0x200000)

(Caution)

- **The maximum size of OS image file can not over 32MB**
- **The maximum size of Logo graphic file can not over 1MB**



<Figure 1-1 >

1.2. The Functions of the MagicLoader

- Function as the second bootloader
- Function as a USB downloader (can download OS image file from PC to SDRAM(or NAND flash) through USB 1.1 device port)
- Function as a diagnostic test program
- Function as a Logo graphic file loader

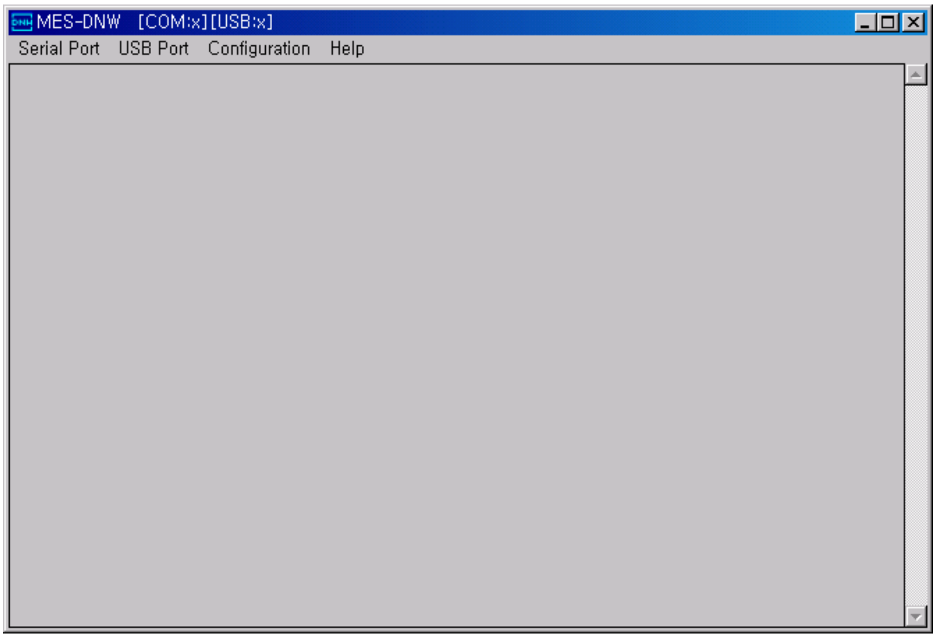
1.3. The Preparations for using the MagicLoader

- Compile Tool : ARM Developer Suite V1.2
- USB cable
- UART cable
- DNW Terminal program
- USB driver for DNW

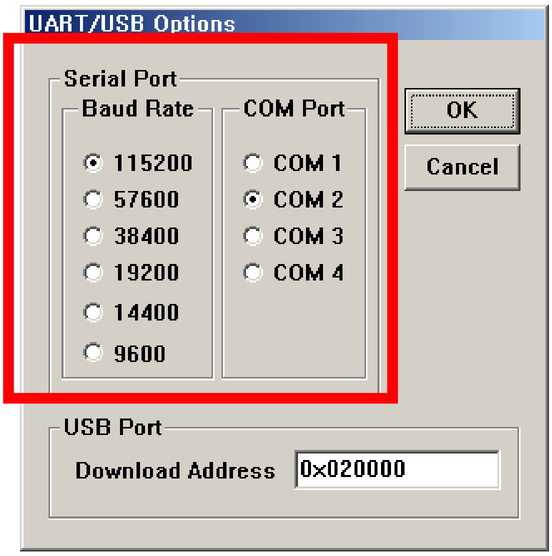
2. The Usage of the MagicLoader

(1) Connect the Serial Cable(UART Cable) between PC and KIT

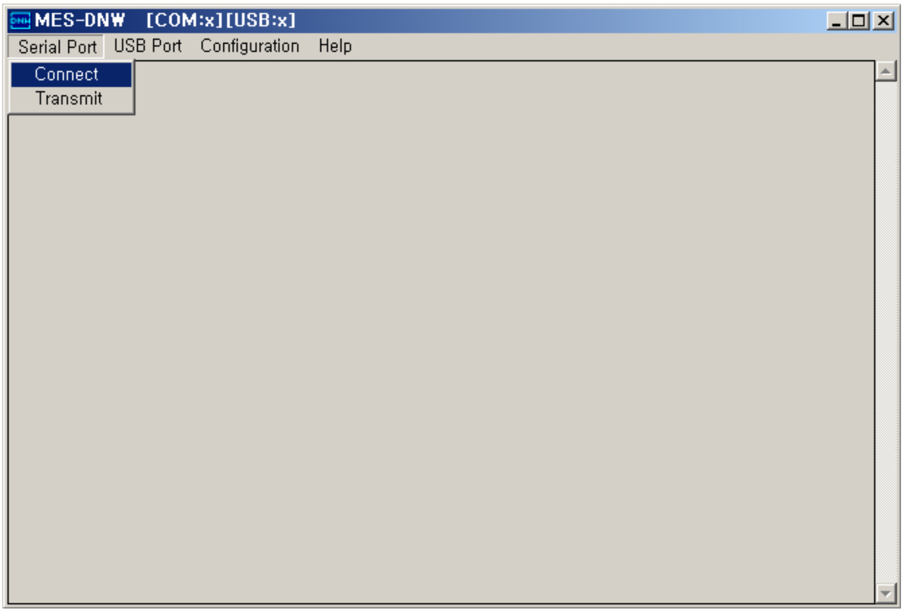
(2) Execute the DNW Terminal



(3) Select **Configuration** → **Option** and set the COM port of PC and Baud rate

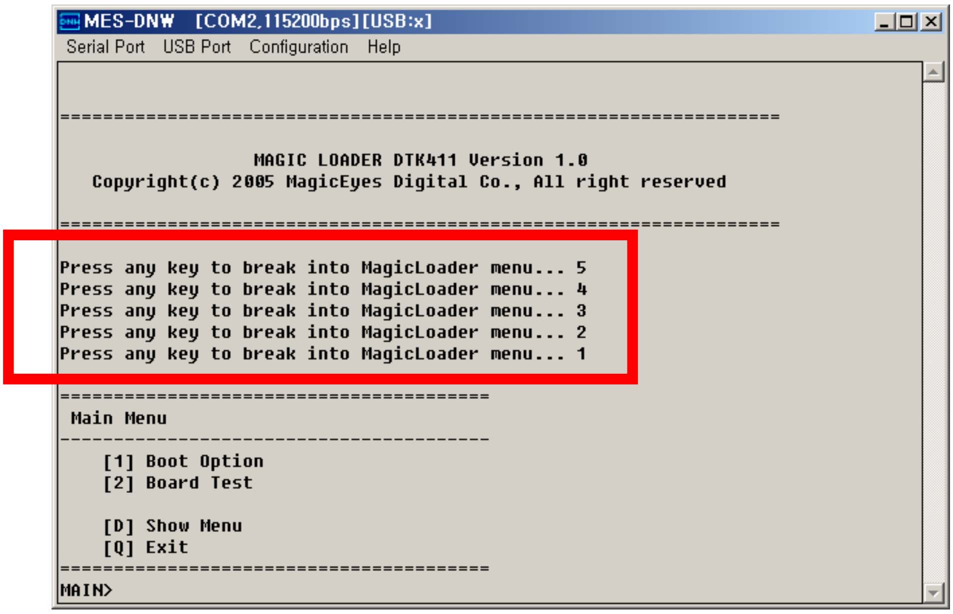


(4) Select **Serial Port** → **Connect**



(5) Turn on the KIT Power

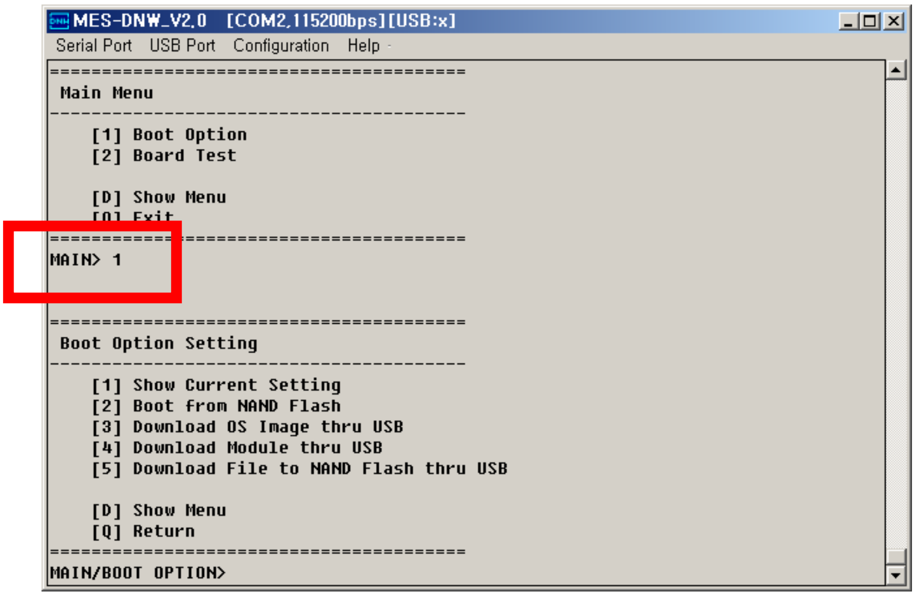
(6) If the MagicLoader starts, you can see the following message, 'Press any key to break into MagicLoader menu…' for 5 seconds. To change the Menu of MagicLoader, press any key of Keyboard. If there is no user's input for 5 seconds, MagicLoader will run according to the setting value which is stored finally.



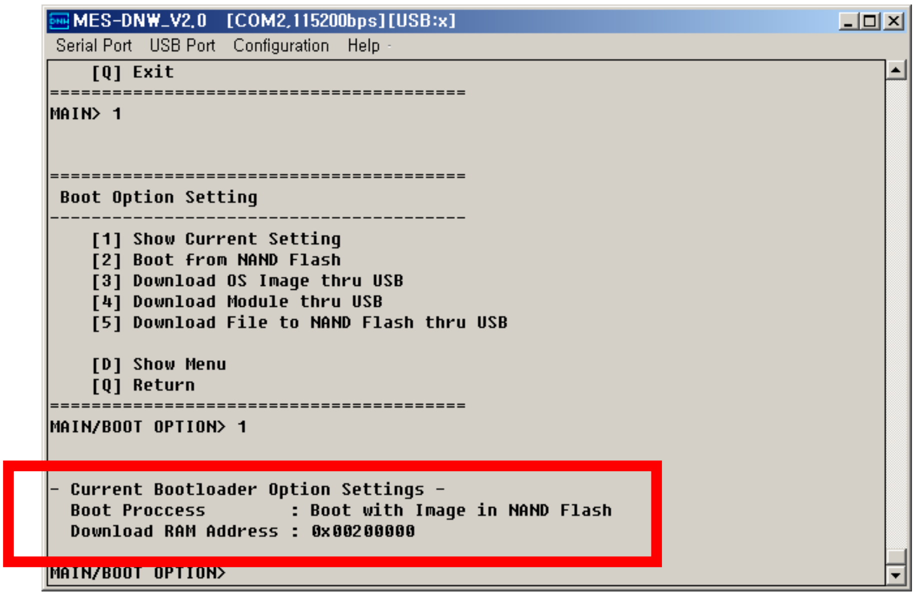
(7) Description of **Main Menu**

- [1] **Boot Option** : User can select Boot Option among the **Boot Option Setting** menus.
- [2] **Board Test** : User can test several functions of KIT.
- [D] **Show Menu** : User can see the Menu again.
- [Q] **Exit** : Escape from **Main Menu** settings.

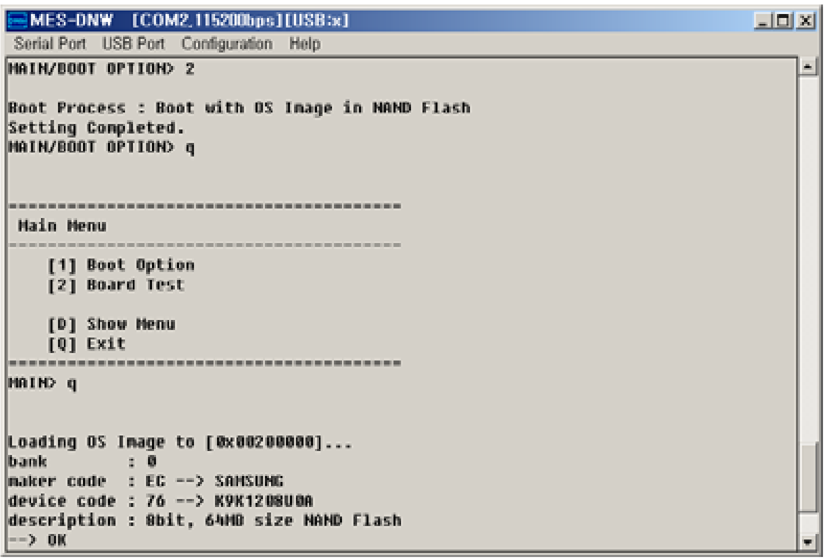
(8) To enter the **Boot Option**, type 1 and press Enter key of Keyboard



(9) [1] **Show Current Setting** : User can see the current setting value of Boot Option



(10) [2] **Boot from NAND Flash** : If user set this option, KIT will be boot with OS image in NAND Flash



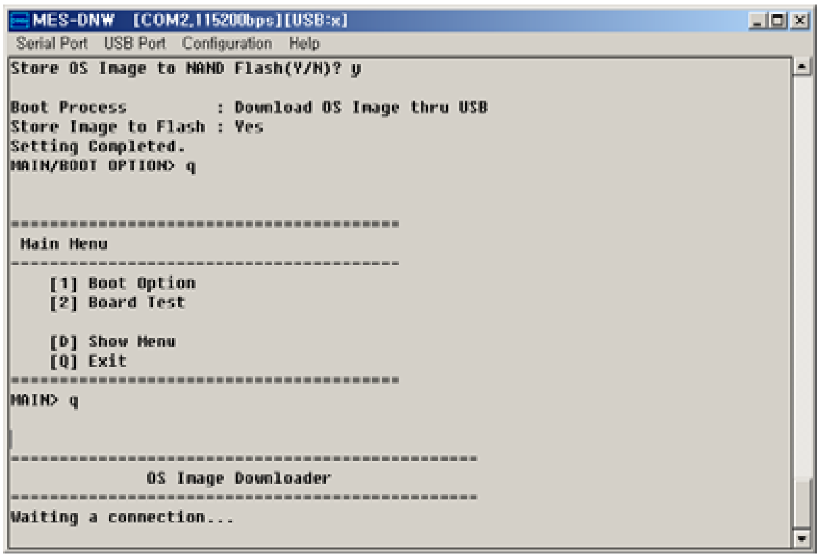
Type q and press Enter key to escape from **Boot Option Setting**

Type q and press Enter key to escape from **Main Menu**

(11) [3] **Download OS Image thru USB** : User can download OS image file through USB1.1 device port. There are two options.

Y : Store OS image file to NAND flash

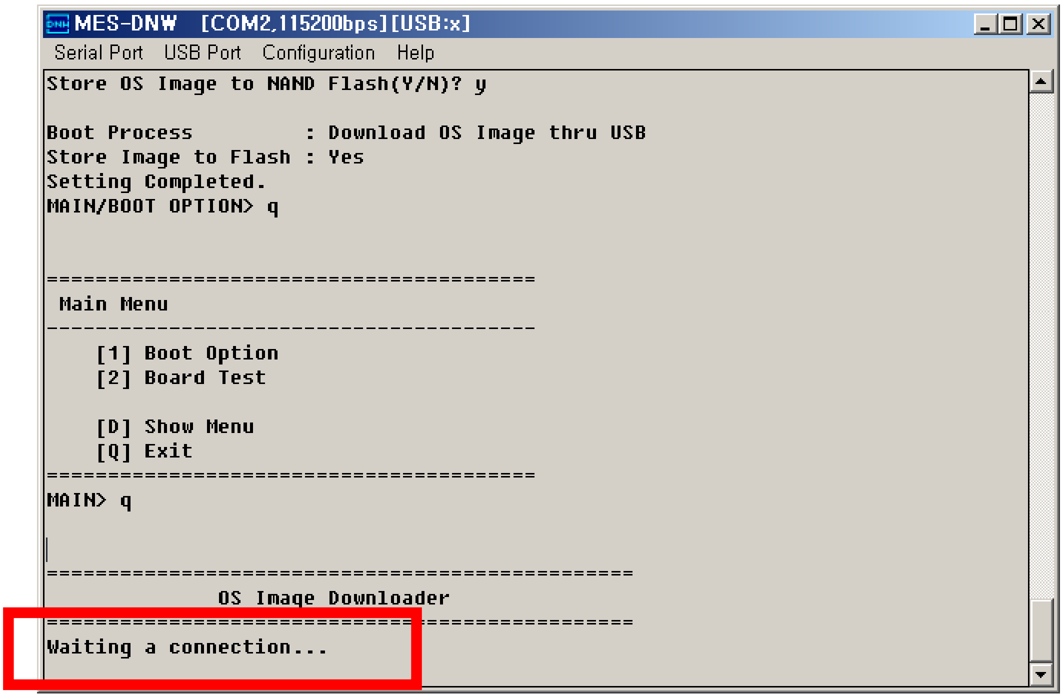
N : Do not store OS image file to NAND flash



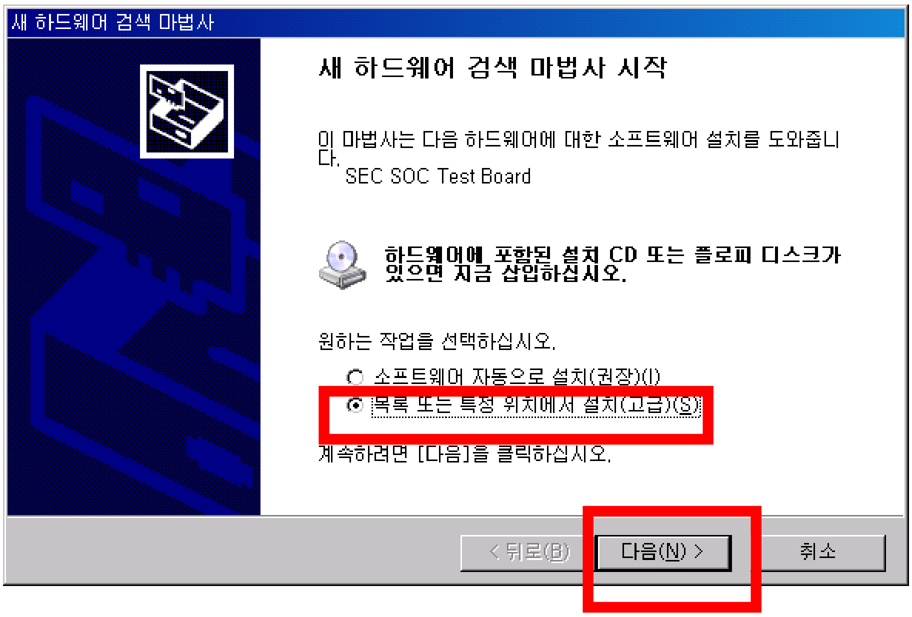
Type q and press Enter key to escape from **Boot Option Setting**

Type q and press Enter key to escape from **Main Menu**

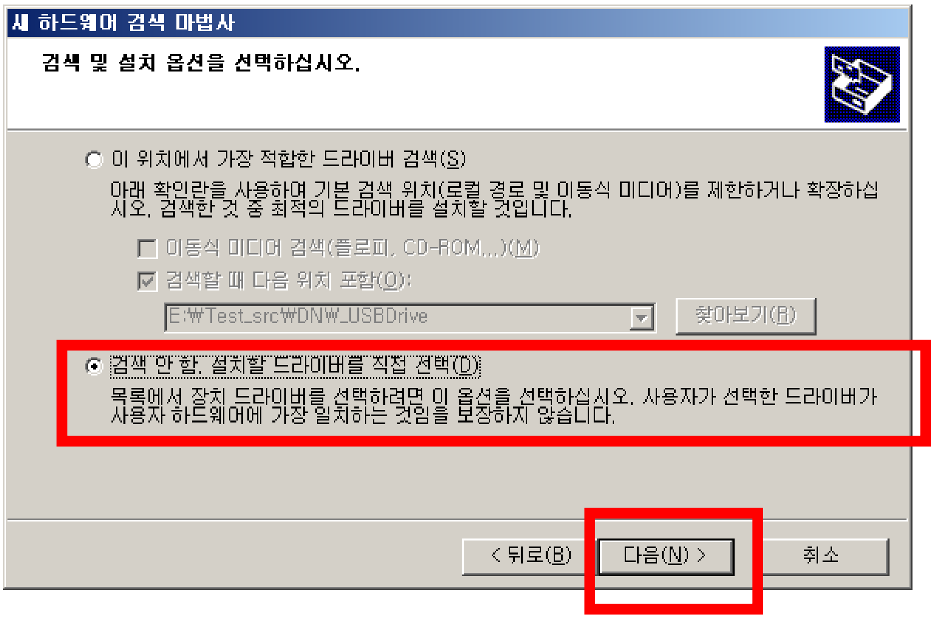
(11-1) ‘Waiting a connection…’ message will be printed in DNW terminal. Connect the USB cable between PC and KIT.



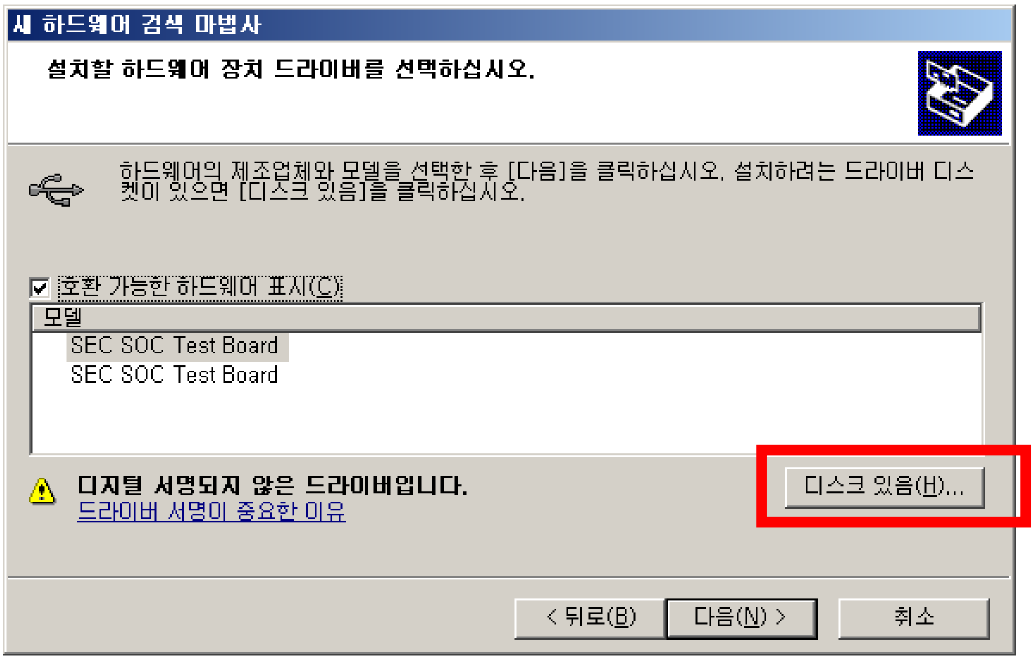
(11-2) PC will detect new hardware and ‘New Hardware Searching Wizard’ window will be popped up. There will be a two options. Do not select ‘auto installation’ option, select ‘installation from list’ option and click ‘Next’ button.



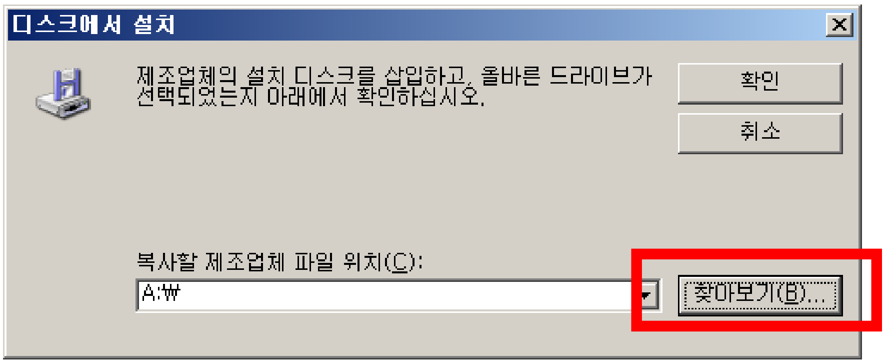
(11-3) Select 'Not search. Select driver directly' option and click 'Next' button



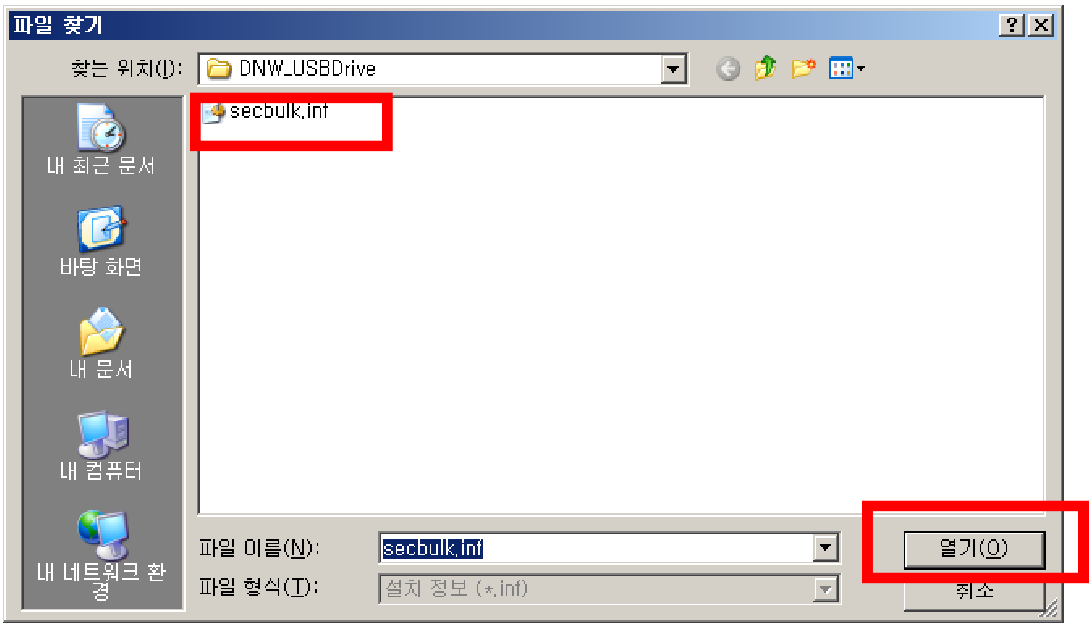
(11-4) Click 'Disk' button



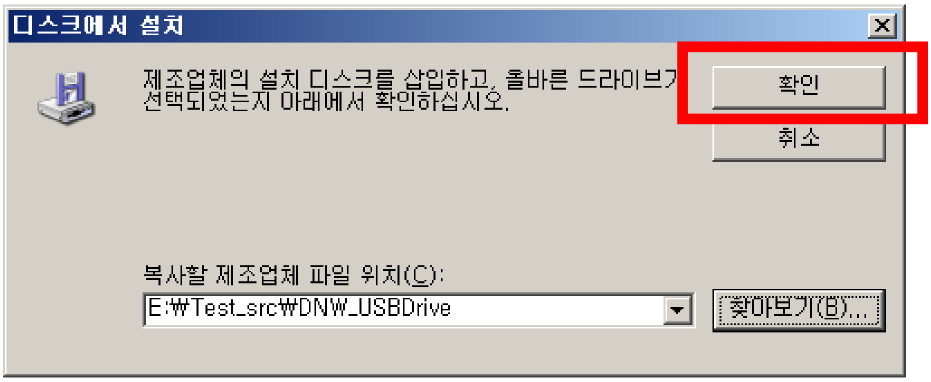
(11-5) Click ‘Search’ button.



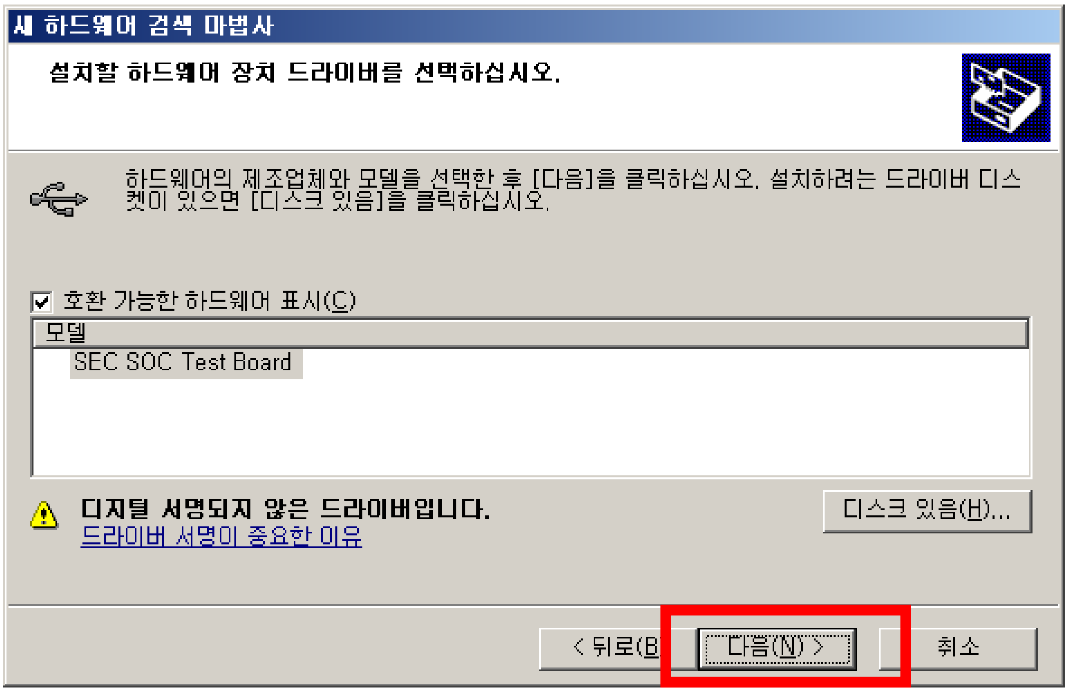
(11-6) Select ‘secbulk.inf’ file and open it.



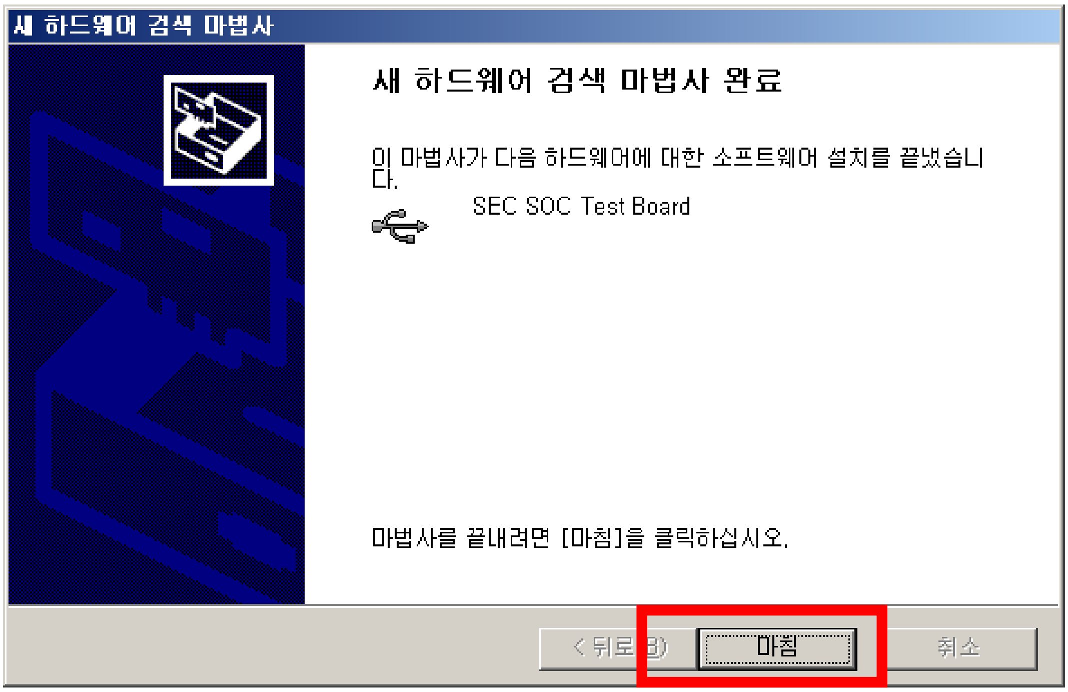
(11-7) Click ‘Confirm’(or OK) button



(11-8) Click ‘Next’ button



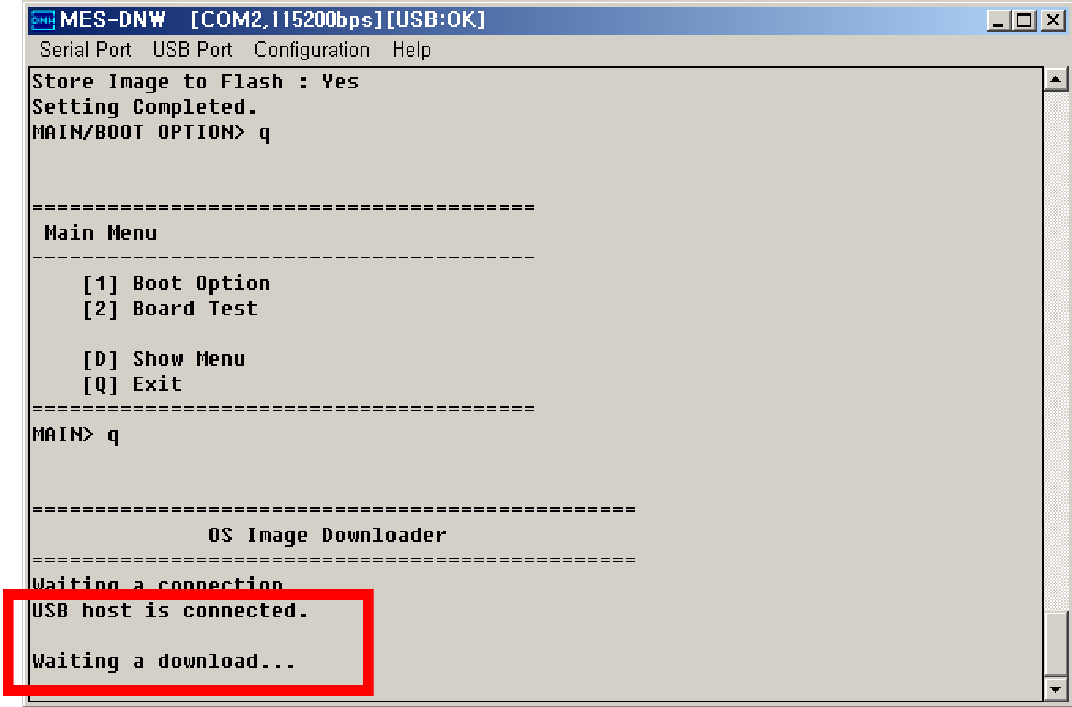
(11-9) Click ‘Finish’ button



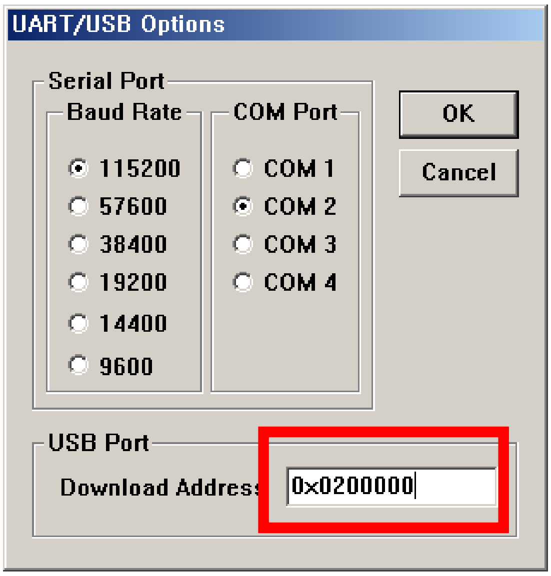
(11-10) If the driver installation is finished successfully, you can see the below message :

'USB host is connected.'

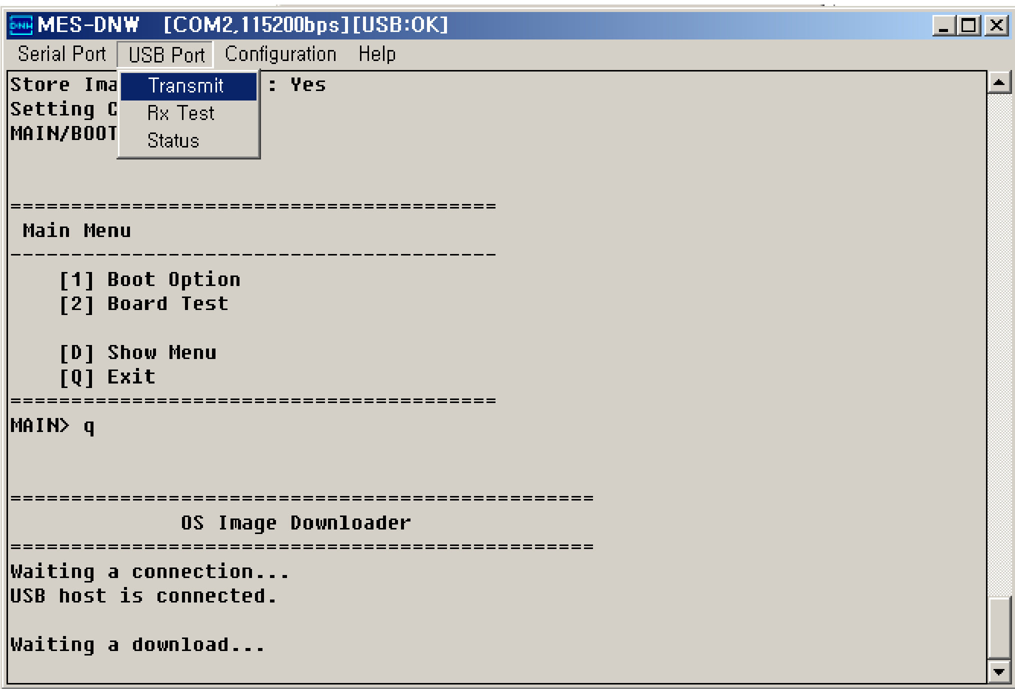
'Waiting a download…'



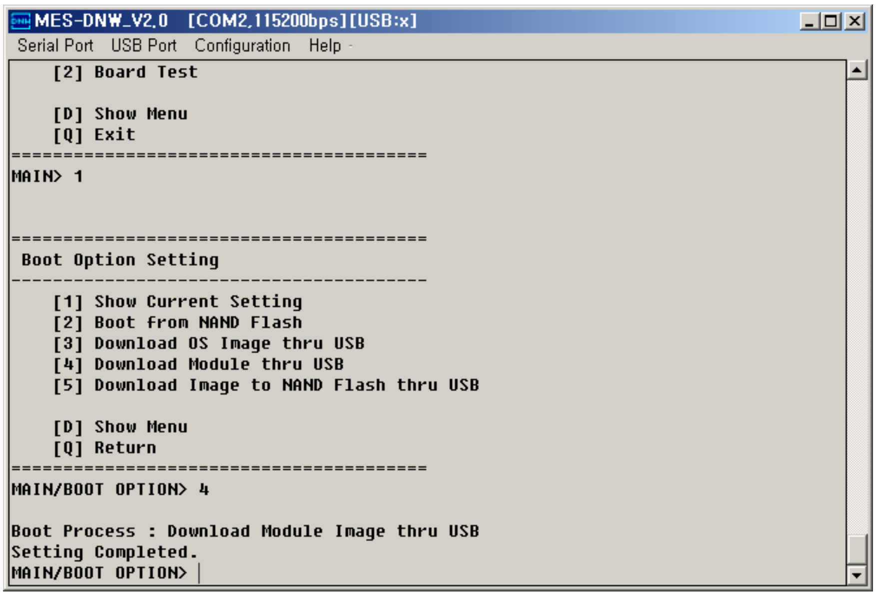
(11-11) Click **Configuration** → **Option** and set **Download Address** to 0x200000.



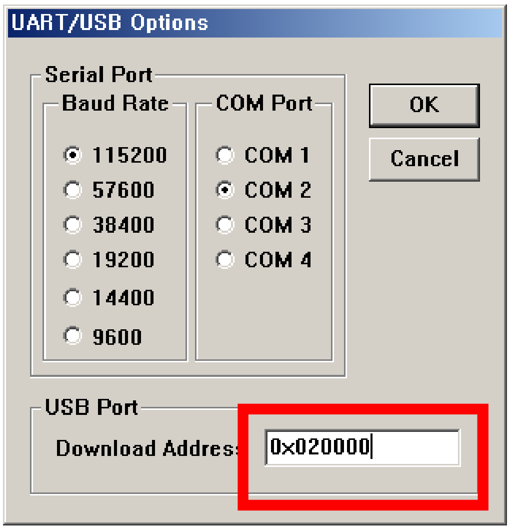
(11-12) Click **USB Port** → **Transmit** and select the OS Image file to download.



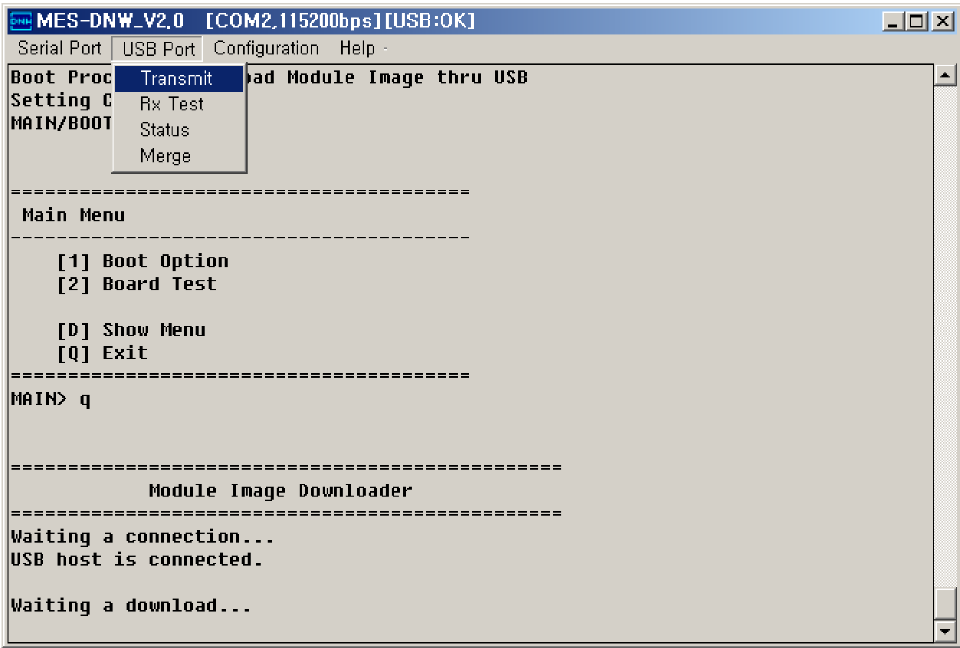
(12) [4] **Download Module Image thru USB** : User can download project image file(ex. Magicloader image file) from PC to SDRAM.



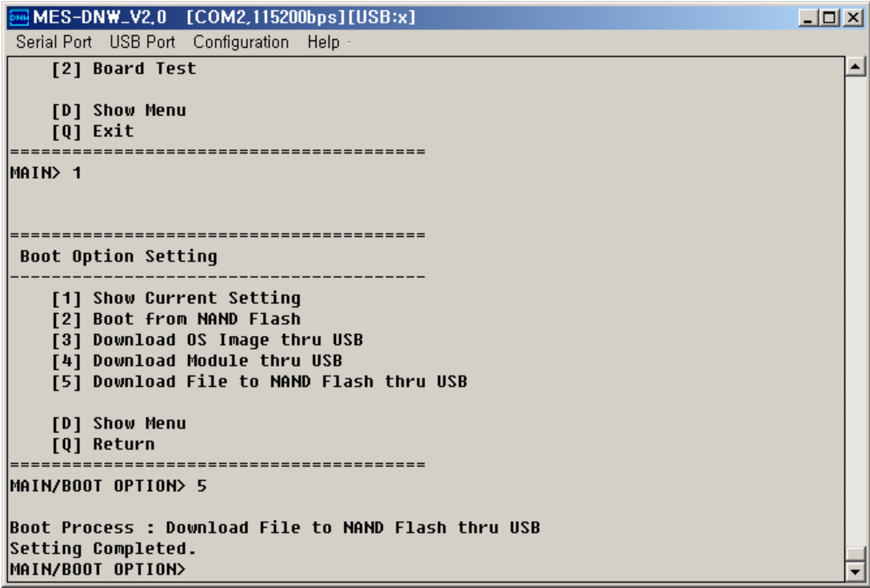
(12-1) Click **Configuration** → **Option** and set the **Download Address** to 0x20000



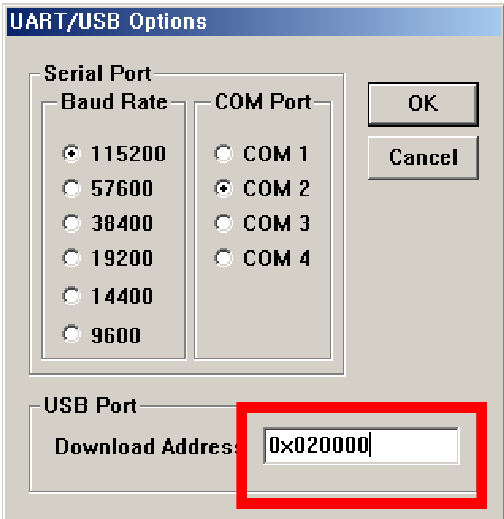
(12-2) Click **USB Port** → **Transmit** and select Project Image file to download.



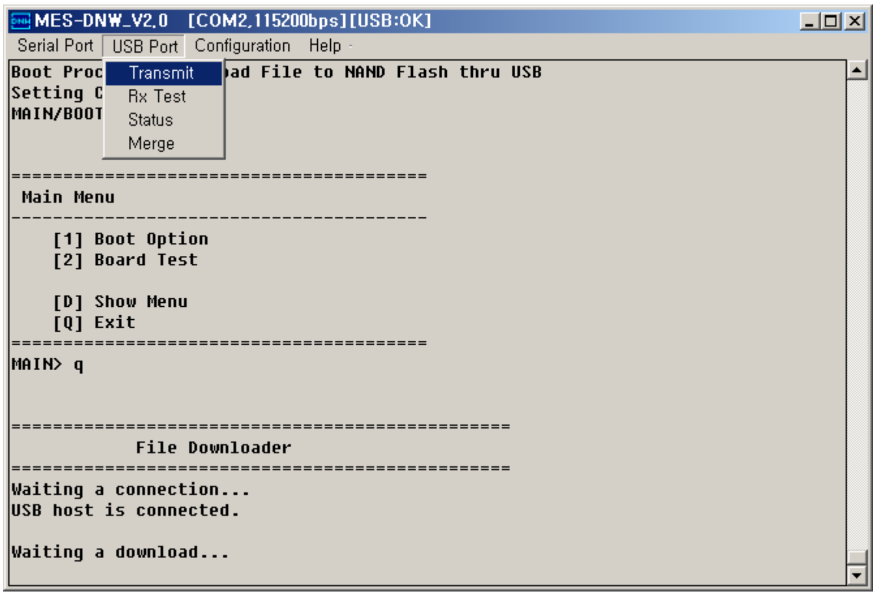
(13)[5] **Download file to NAND Flash thru USB** : User can download any kind of file from PC to NAND Flash



(13-1) Click **Configuration** → **Option** and set the **Download Address** of NAND Flash



(13-2) Click USB **Port** → **Transmit** and select Image file to download.



3. How to display logo graphic file

3.1. Make logo graphic file

- Logo graphic file must be 16bit RGB DATA Format or 24bit RGB DATA Format

- BMP2DMP.exe is a file format converter from 24BIT BMP file to 16bit RGB DATA Format and from 24BIT BMP file to 24t RGB DATA Format

(1) Commnad execute and bmp2dmp execute

Usage : bmp2dmp srcfilename dstfilename <RGB DATA>

```
명령 프롬프트
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

D:₩>bmp2dmp
Usage : bmp2dmp.exe srcfilename dstfilename RGB_DATA
<<RGB DATA>>
0 : 24Bit BMP File --> 16Bit RGB DATA
1 : 24Bit BMP File --> 24Bit RGB DATA
```

(2) 24Bit BMP File → 16Bit RGB DATA (ex:24bit 800x480 BMP file)

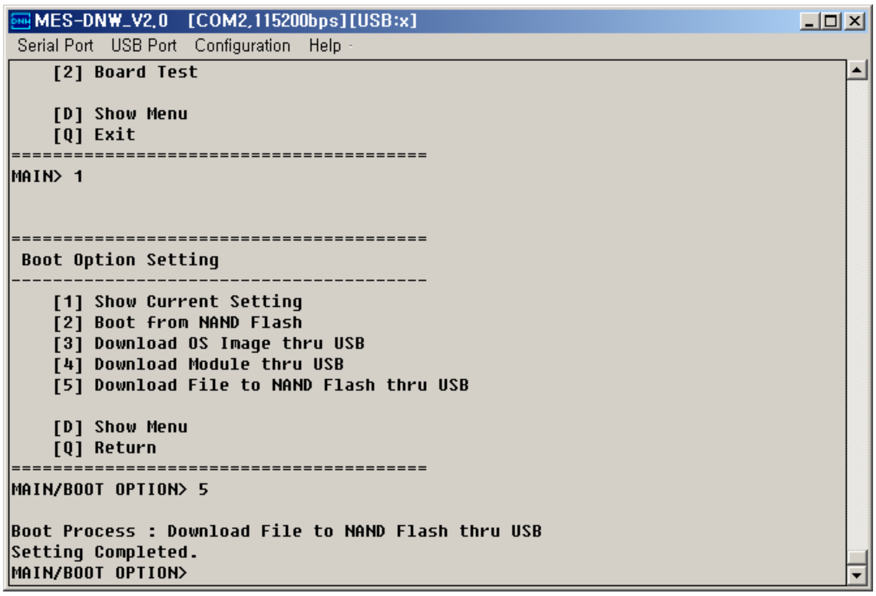
```
명령 프롬프트
D:₩>bmp2dmp Car_800x480.bmp Car_800x480.bin 0
Width : 800, Height : 480
16BIT RGBDATA
```

(3) 24Bit BMP File → 24Bit RGB DATA (ex:24bit 800x480 BMP file)

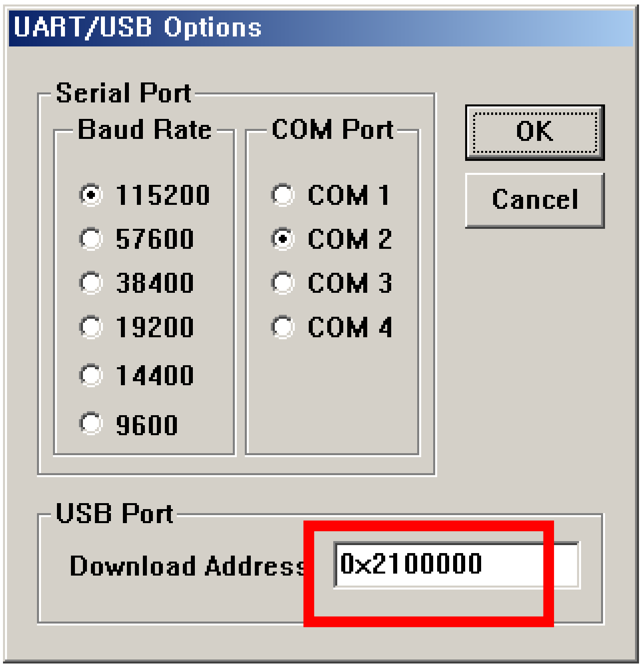
```
명령 프롬프트
D:₩>bmp2dmp Car_800x480.bmp Car_800x480.bin 1
Width : 800, Height : 480
24BIT RGBDATA
```

3.2. Write logo graphic file to NAND Flash

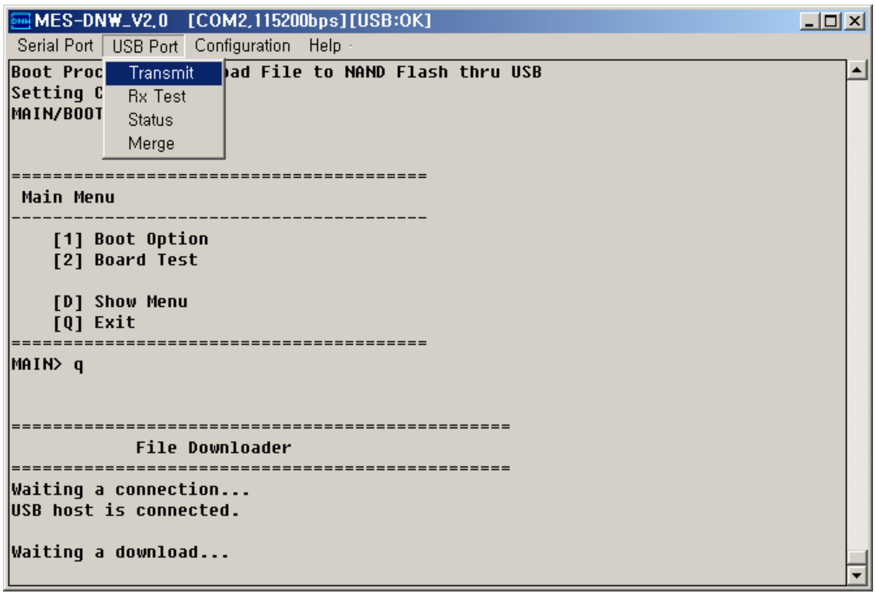
(1) Select ‘[5] Download file to NAND Flash thru USB ‘ of Boot option Setting menu



(2) Click Configuration → Option and set the Logo Address of NAND Flash (0x2100000)



(3) Click **USB Port → Transmit** and select logo graphic file to download.



3.3. Display Device Configuration

(1) Default Display Configuration Value is for Hitachi 7” Wide VGA LCD of DTK411

(2) User can modify the Display Configuration Value for user’s LCD. To do this, open the ‘OEM_MMSP2_SYSTEM_DTK411.h’ file and add or modify the LCD Display Configuration Value, like as Horizontal width, Vertical width, clock timing, and so on

3.3.1. Logo image display function - parameter setting examples for 7inch & 4.3inch LCD

MagicLoader V2.2 supports the logo-display function and includes the parameter setting example for Hitachi 7” wide VGA LCD and Samsung 4.3” wide QVGA LCD.

The OEM_MMSP2_SYSTEM_DTK411.h includes the parameter setting values for these LCDs.

- Display Clock

```
// Hitachi wide 800x480 7”LCD
#define VID_DISPCLKSOURCE          VID_DISPCLK_SOURCE_APLL
#define VID_DISPCLKDIVIDE          4
#define VID_DISPCLKPOL             FALSE

// Samsung 480x272 4.3"LCD
#define VID_DISPCLKSOURCE          VID_DISPCLK_SOURCE_FPLL
#define VID_DISPCLKDIVIDE          21
#define VID_DISPCLKPOL             FALSE
```

- Display order

```
// Hitachi wide 800x480 7"LCD
#define DPC_DISPODER               DPC_RGB_666

// Samsung 480x272 4.3"LCD
#define DPC_DISPODER               DPC_RGB_888
```

- Display Resolution

```
// Hitachi wide 800x480 7"LCD
#define VID_MAX_X_RESOLUTION       800
#define VID_MAX_Y_RESOLUTION       480

// Samsung 480x272 4.3"LCD
#define VID_MAX_X_RESOLUTION       480
#define VID_MAX_Y_RESOLUTION       272
```

■ Display SYNC

```
// Hitachi wide 800x480 7"LCD
#define VID_HSYNC_WIDTH             21
#define VID_HSYNC_FRONT_PORCH       40
#define VID_HSYNC_BACK_PORCH        88
#define VID_HSYNC_ACTIVEHIGH        POLARITY_ACTIVELOW
#define VID_VSYNC_WIDTH             2
#define VID_VSYNC_FRONT_PORCH       11
#define VID_VSYNC_BACK_PORCH        25
#define VID_VSYNC_ACTIVEHIGH        POLARITY_ACTIVELOW

// Samsung 480x272 4.3"LCD
#define VID_HSYNC_WIDTH             41
#define VID_HSYNC_FRONT_PORCH       2
#define VID_HSYNC_BACK_PORCH        2
#define VID_HSYNC_ACTIVEHIGH        POLARITY_ACTIVEHIGH
#define VID_VSYNC_WIDTH             10
#define VID_VSYNC_FRONT_PORCH       2
#define VID_VSYNC_BACK_PORCH        2
#define VID_VSYNC_ACTIVEHIGH        POLARITY_ACTIVEHIGH
```

■ Display Clock Polarity

```
// Hitachi wide 800x480 7"LCD
#define VID_CLKH_POLARITY           TRUE

// Samsung 480x272 4.3"LCD
#define VID_CLKH_POLARITY           TRUE
```

■ MLC RGB Bit Per Pixel

```
// Hitachi wide 800x480 7"LCD
#define MLC_RGB_BPP                 MLC_RGB_16BPP

// Samsung 480x272 4.3"LCD
#define MLC_RGB_BPP                 MLC_RGB_24BPP
```

3.3.2. Progress-bar function

MagicLoader V2.2 supports the progress-bar graphic function. It can be used for progress-bar display during CPU reads the OS image file from NAND Flash to main memory(DRAM).

OEM_MMSP2_SYSTEM_DTK411.h includes the parameters related to X/Y Position, Width, and Height of progress-bar.

```
// Display Bar
#define DISPLAY_BAR_START_X             VID_MAX_X_RESOLUTION/4
#define DISPLAY_BAR_START_Y             (VID_MAX_Y_RESOLUTION*3)/4
#define DISPLAY_BAR_WIDTH               VID_MAX_X_RESOLUTION/2
#define DISPLAY_BAR_HEIGHT              20
```

DrawWall() and PutStringXY() function in Main.c draw a quadrangle outline and a text, respectively.

The main.c from file the DrawWall function shows a quadrangular outline in the LCD and the PutStringXY function shows the Text in the LCD.

```
DrawWall(DISPLAY_BAR_START_X, DISPLAY_BAR_START_Y, DISPLAY_BAR_WIDTH, DISPLAY_BAR_HEIGHT);

PutStringXY( DISPLAY_BAR_START_X,                        // X position
            (DISPLAY_BAR_START_Y-DISPLAY_BAR_HEIGHT),    // Y position
            COLOR_WHITE,                                 // Foreground Color
            COLOR_BLACK,                                 // Background Color
            (const SCHAR*)"Loading OS Image..." );       // String
```

DrawVLine() function in Flash.c draws the progress status of progress-bar according to the speed of OS image loading from NAND Flash to main memory.

3.3.3. Logo Image store address set to NAND Flash

User can modify the size & the storage address of logo image file in NAND flash according to user's LCD . The size & the storage address of logo image file can be modified in OEM_MMSP2_SYSTEM_DTK411.h.

```
// Display Bar
#define LOGO_IMAGE_FLASH_START_PAGE        67584      // 0x2100000/PAGE SIZE(512)
#define LOGO_IMAGE_FLASH_SIZE              0x00100000 // 1M
```